SLIC Index V3

Methodology Technical Paper

English master edition

Published snapshot: 2026-04-23T17:19:44.917Z

This paper documents the live published SLIC scorer as implemented in **scripts/rescore-all-cities.mjs** and written to **src/data/publishedRankingData.json**. It replaces earlier draft descriptions of a 35-indicator anchor model that no longer matches the public dataset.

Snapshot fact	Value
Cities in public file	163
Ranked cities	154
Watchlist cities	9
Scored metric lines	20
Visible non-scoring diagnostics	3

1. Published method

SLIC has five public pillars. Within each pillar, observed scored metrics are combined by weighted mean. Across pillars, the final score uses a weighted AMPI, so cross-pillar imbalance is punished mathematically rather than rhetorically.

```
normalize(x, p05, p95, positive) = 100 * clamp((winsor(x) - p05) / (p95 - p05), 0, 1)
normalize(x, p05, p95, negative) = 100 * clamp((p95 - winsor(x)) / (p95 - p05), 0, 1)

P_p(c) = sum observed alpha_(p,m) * q_m(c) / sum observed alpha_(p,m)

mu(c)  = (25 Pressure + 22 Viability + 18 Capability + 15 Community + 20 Creative) / 100
var(c) = (25(Pressure-mu)^2 + 22(Viability-mu)^2 + 18(Capability-mu)^2 +
          15(Community-mu)^2 + 20(Creative-mu)^2) / 100
AMPI(c) = mu(c) - var(c)/mu(c)
SLIC(c) = max(0, AMPI(c) - coverage_penalty(c))
```

The final published score is therefore **not** a simple weighted average of the displayed pillar scores. That distinction matters: a city with one catastrophic pillar is supposed to score worse than a city with the same weighted mean but more even performance.

2. What is scored

The current public model scores **20** metric lines and keeps **3** additional metric lines visible as diagnostics only. Visibility and scoring are intentionally separated.

Growth (25)

Metric	Weight	Status	Method note
Tax-adjusted PPP disposable income	9	Scored	Residual money after essentials, converted once through the PPP private-consumption factor.
Housing burden	5	Scored	Housing cost share of income; higher burden scores worse.
Household debt burden	4	Scored	Debt relative to income, with fallback when city debt evidence is thin.
Working time pressure	4	Scored	Higher work burden scores worse.
Suicide and severe mental strain	3	Scored	Public-health strain proxy; higher severe-strain burden scores worse.
Economic growth momentum	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Viability (22)

Metric	Weight	Status	Method note
Personal safety	5	Scored	Harm and victimization outcomes, not surveillance theatre.
Transit access and commute burden	5	Scored	Better access and lower burden score better.
Clean air	4	Scored	Higher pollution exposure scores worse.
Water, sanitation, and utility reliability	4	Scored	Safer access and higher reliability score better.
Digital infrastructure	4	Scored	Broadband quality and affordability.
Climate and sunlight livability	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Capability (18)

Metric	Weight	Status	Method note
Healthcare quality	8	Scored	Effective-care quality and access.
Education quality	6	Scored	Learning quality and skills formation.

Metric	Weight	Status	Method note
Equal opportunity and distributional fairness	4	Scored	Composite scored metric: 70% equal-opportunity evidence, 30% reversed Gini context.

Community (15)

Metric	Weight	Status	Method note
Hospitality and belonging	5	Scored	Do people feel welcome in practice?
Tolerance and pluralism	5	Scored	Composite scored metric: 40% Equaldex, 30% Freedom House, 30% reversed hate-crime or civil-liberties proxy.
Cultural, historic, and public-life vitality	5	Scored	Everyday public-life texture; visitor demand is contextual, not an automatic bonus.
Birth-rate optimism	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Creative (20)

Metric	Weight	Status	Method note
Entrepreneurial dynamism	6	Scored	Startup formation and productive entry.
Innovation and research intensity	5	Scored	Research depth and knowledge production.
Economic vitality and productive context	5	Scored	Composite scored metric: 50% investment signal, 30% GDP per capita PPP, 20% GDP growth.
Administrative and investment friction	4	Scored	Higher time-cost and licensing burden score worse.

3. Composite metrics

Three scored terms are composites. When one component is missing, the composite is reweighted over the observed component weights only; nothing is silently imputed.

```
EqualOpportunityFairness = 0.7 * EqualOpportunity + 0.3 * ReverseGini
TolerancePluralism      = 0.4 * Equaldex + 0.3 * FreedomHouse + 0.3 * ReverseHateCrime
EconomicVitality        = 0.5 * InvestmentSignal + 0.3 * GDPperCapitaPPP + 0.2 * GDPGrowth
```

4. Coverage, watchlist logic, and ranking protocol

```
cov_direct = 1 if observed else 0
cov_composite = observed component weight / total component weight

Cov_p(c) = sum alpha_(p,m) * cov_m(c) / sum alpha_(p,m)
Cov(c)   = (25 Cov_pressure + 22 Cov_viability + 18 Cov_capability +
            15 Cov_community + 20 Cov_creative) / 100
```

Coverage grade	Rule	Penalty	Rank status
A	Cov >= 0.75	0	Ranked
B	0.50 <= Cov < 0.75	5	Ranked
C	0.35 <= Cov < 0.50	15	Ranked
Watchlist	Cov < 0.35 or manual watchlist override	0	Watchlist

After score computation, ranked cities are placed into 10-city buckets with a one-per-region rule. Taiwan is exempt from the region uniqueness check, and the Alpha tier also applies liveability floors of Community >= 40 and Growth >= 40. This means displayed ordinal rank is score-first, then filtered through the published bucket protocol.

5. Worked example from the live dataset

Example city: **Kaohsiung**. This walkthrough uses the published row in the live dataset, not a fictional demonstration.

```
Negative-direction normalization example (personal safety)
raw = 0.8
p05 = 0.23
p95 = 22.63
score = 100 * (22.63 - 0.8) / (22.63 - 0.23)
score = 97.5

Growth weighted mean
Growth = (9x81.5 + 5x81.1 + 4x41.4 + 3x27.5) / 21
Growth = 66.1

Cross-pillar AMPI
mu  = 70.308
var = 252.971
AMPI = 70.308 - 252.971 / 70.308
AMPI = 66.710
Coverage = 0.77 -> grade A -> penalty 0
Published SLIC = 66.7
```

The important methodological point is visible in the arithmetic: the weighted pillar mean for this city is higher than the final score because AMPI makes uneven pillar structure costly.

6. Data profile of the current public file

Measure	Value
City / metro metric share	48.7%
National / international metric share	25.9%
Composite metric share	15.7%
Derived metric share	9.6%

Measure	Value
Grade A cities	138
Grade B cities	2
Grade C cities	16
Watchlist cities	9

These shares are computed directly from non-null metric lines in the published JSON. They are included here so the paper does not rely on invented methodology graphics or unsupported source-mix claims.

7. Published ranked cities

The table below is the ranked slice of the current published dataset. Watchlist cities are excluded from this list by definition.

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
1	Raleigh	United States	69.0	75.1	76.0	79.1	53.8	62.0	A
2	Montreal	Canada	68.9	62.3	93.3	75.3	74.1	54.1	A
3	Kaohsiung	Taiwan	66.7	66.1	86.5	90.6	47.1	56.9	A
4	Eindhoven	Netherlands	64.3	64.6	97.6	71.8	65.7	42.8	A
5	Taipei	Taiwan	63.8	48.4	89.5	89.7	47.1	69.4	A
6	Prague	Czechia	63.5	62.0	90.6	61.9	87.7	42.4	A
7	Jeju City	South Korea	61.6	56.2	99.2	85.8	44.8	50.6	A
8	Perth	Australia	59.8	45.3	89.5	87.1	56.3	51.2	A
9	Valparaiso	Chile	58.9	58.3	72.7	73.6	56.9	43.0	A
10	Bangkok	Thailand	58.5	45.4	84.9	62.5	90.0	46.0	A
11	Minneapolis	United States	64.6	71.6	69.1	79.1	44.2	62.0	A
12	Graz	Austria	63.9	62.4	87.1	74.4	73.4	41.4	A
13	Busan	South Korea	60.5	53.2	99.2	85.8	44.8	50.6	A
14	Singapore	Singapore	60.2	43.3	87.4	94.1	38.8	73.3	A
15	Adelaide	Australia	58.5	43.7	89.5	87.1	56.3	49.2	A
16	Santiago	Chile	57.9	50.7	72.7	83.8	56.9	44.6	A
17	Torun	Poland	56.0	68.4	79.5	65.1	52.0	31.9	A
18	Abu Dhabi	United Arab Emirates	55.9	70.5	92.6	62.0	29.5	46.2	A
19	Port Louis	Mauritius	51.2	56.0	76.2	47.0	39.6	45.4	A
20	Male	Maldives	45.0	54.6	72.1	39.8	35.8	35.4	A
21	Pittsburgh	United States	64.5	76.0	65.1	79.1	50.2	55.7	A
22	Lyon	France	60.6	52.4	97.9	63.9	73.1	45.1	A
23	Incheon	South Korea	60.5	53.3	99.2	85.8	44.8	50.6	A
24	Brisbane	Australia	58.2	39.7	89.5	87.1	56.3	54.4	A
25	Dubai	United Arab Emirates	55.8	69.8	92.6	62.0	29.5	46.2	A
26	Katowice	Poland	55.4	65.9	79.5	65.1	52.0	31.9	A
27	Melaka	Malaysia	49.5	63.0	75.3	52.4	40.3	31.5	A
28	Montevideo	Uruguay	49.1	52.4	59.5	74.6	64.6	24.4	A
29	Gaborone	Botswana	35.7	51.4	31.9	22.2	45.2	39.0	A
30	Hyderabad	India	31.3	44.7	53.1	30.4	33.5	14.6	A
31	Ottawa	Canada	63.5	55.3	93.3	75.3	75.6	44.2	A
32	Braga	Portugal	60.6	58.4	95.7	68.8	57.9	43.8	A
33	Suwon	South Korea	60.3	53.3	93.1	85.8	45.1	50.6	A
34	Melbourne	Australia	58.2	36.4	89.5	87.1	56.3	60.5	A
35	Manama	Bahrain	55.3	66.4	79.7	57.8	39.7	42.5	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
36	Gdansk	Poland	54.3	60.9	79.5	65.1	52.0	31.9	A
37	Kota Kinabalu	Malaysia	49.3	61.5	77.9	52.4	40.0	31.5	A
38	Cordoba	Argentina	44.5	67.5	69.2	78.6	46.8	10.2	A
39	Windhoek	Namibia	32.9	42.7	30.5	32.1	39.3	25.4	A
40	Pune	India	31.2	44.4	53.1	30.4	33.5	14.6	A
41	Milan	Italy	60.5	88.5	85.8	66.9	58.4	29.4	A
42	Tokyo	Japan	59.5	46.2	94.2	76.8	70.2	44.2	A
43	Toronto	Canada	59.3	38.7	93.3	75.3	72.5	54.1	A
44	Dunedin	New Zealand	57.3	51.2	92.6	85.7	57.2	36.5	A
45	Bratislava	Slovakia	53.9	57.3	89.7	49.7	63.2	35.3	A
46	Kuching	Malaysia	49.3	61.5	73.7	52.4	41.1	31.5	A
47	Khobar	Saudi Arabia	48.9	91.3	96.3	78.9	17.2	24.1	A
48	Curitiba	Brazil	42.2	51.8	45.1	69.5	36.3	27.6	A
49	Rabat	Morocco	32.4	46.9	59.9	34.1	24.2	18.8	A
50	Bengaluru	India	31.1	43.9	53.1	30.4	33.5	14.6	A
51	Copenhagen	Denmark	59.9	33.1	97.9	71.3	89.6	61.5	A
52	Chicago	United States	58.6	56.0	56.6	79.1	47.6	62.0	A
53	Fukuoka	Japan	57.6	59.0	94.5	67.4	46.6	42.3	A
54	Christchurch	New Zealand	57.0	45.8	92.6	85.7	57.2	41.4	A
55	Krakow	Poland	52.3	53.5	79.5	65.1	52.0	31.9	A
56	Jeddah	Saudi Arabia	48.8	90.6	96.3	78.9	17.2	24.1	A
57	George Town	Malaysia	48.6	58.0	73.7	52.4	41.1	31.5	A
58	Florianopolis	Brazil	41.9	50.4	45.1	69.5	36.3	27.6	A
59	Casablanca	Morocco	32.3	46.0	59.9	34.1	24.2	18.8	A
60	Chandigarh	India	30.6	43.9	44.1	30.4	34.1	14.6	A
61	Porto	Portugal	58.6	52.5	95.7	68.8	57.9	43.8	A
62	New York	United States	57.7	27.1	85.4	82.0	68.5	75.3	A
63	Wellington	New Zealand	55.9	37.8	92.6	85.7	57.2	48.8	A
64	Hiroshima	Japan	55.8	59.0	94.5	67.4	46.6	37.3	A
65	Bucharest	Romania	49.5	57.5	81.4	50.6	40.7	34.8	A
66	Riyadh	Saudi Arabia	48.7	89.9	96.3	78.9	17.2	24.1	A
67	Kuala Lumpur	Malaysia	46.8	51.0	73.7	52.4	41.1	31.5	A
68	Buenos Aires	Argentina	41.8	55.4	69.2	78.6	46.8	10.2	A
69	Kigali	Rwanda	30.5	56.0	47.2	15.8	30.5	24.8	A
70	Kandy	Sri Lanka	29.0	51.5	40.7	33.7	28.6	11.7	A
71	Zurich	Switzerland	58.3	39.1	97.6	72.7	76.0	49.4	A
72	Vancouver	Canada	57.1	34.7	93.3	75.3	70.2	54.1	A
73	Sapporo	Japan	55.8	59.0	94.5	67.4	46.6	37.3	A
74	Sydney	Australia	54.5	27.8	89.5	87.1	56.3	63.9	A
75	Budapest	Hungary	48.1	47.2	93.1	55.5	63.5	26.2	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
76	Hat Yai	Thailand	46.6	55.1	66.7	56.4	52.2	22.8	A
77	Salalah	Oman	45.4	73.9	71.1	42.0	31.2	30.6	A
78	Sao Paulo	Brazil	39.9	42.7	45.1	69.5	36.3	27.6	A
79	Colombo	Sri Lanka	28.9	48.8	40.7	33.7	28.6	11.7	A
80	Johannesburg	South Africa	26.3	34.3	15.1	26.3	42.3	29.8	A
81	Tallinn	Estonia	58.2	51.0	90.1	58.4	49.0	57.8	A
82	Kobe	Japan	55.4	58.0	94.5	67.4	46.6	37.3	A
83	Auckland	New Zealand	54.4	32.4	92.6	85.7	57.2	52.9	A
84	San Juan	Puerto Rico	48.2	43.8	59.2	68.3	63.3	45.8	B
85	Phuket	Thailand	44.9	48.0	68.0	56.4	52.2	22.8	A
86	Muscat	Oman	44.7	66.9	71.1	42.0	31.2	30.6	A
87	Belgrade	Serbia	43.1	46.5	74.8	53.2	32.1	30.3	A
88	Arequipa	Peru	36.3	57.5	34.1	42.3	34.9	24.7	A
89	Thimphu	Bhutan	28.6	47.2	47.0	25.2	40.4	10.2	A
90	Cape Town	South Africa	26.2	34.2	15.1	26.3	42.3	29.8	A
91	Helsinki	Finland	58.2	40.9	91.5	81.0	57.5	52.9	A
92	Hobart	Australia	52.5	33.8	89.5	87.1	56.3	45.1	A
93	San Jose	Costa Rica	45.3	45.7	43.5	66.0	47.9	35.9	A
94	Doha	Qatar	44.7	59.0	73.9	54.4	18.6	37.6	A
95	Chiang Mai	Thailand	43.8	54.3	49.7	56.4	52.2	22.8	A
96	Tianjin	China	42.4	43.5	98.6	71.9	19.7	37.1	A
97	Vilnius	Lithuania	36.9	65.0	-	41.6	74.0	41.4	C
98	Lima	Peru	35.8	51.3	34.1	42.3	34.9	24.7	A
99	Accra	Ghana	25.3	55.6	35.9	14.9	30.3	17.8	A
100	Pokhara	Nepal	22.3	44.3	37.6	12.2	21.7	16.6	A
101	London	United Kingdom	58.2	35.1	82.8	78.4	70.2	59.3	A
102	Panama City	Panama	44.2	61.0	44.0	64.2	41.8	26.9	A
103	Kuwait City	Kuwait	42.2	91.4	69.2	65.0	19.7	19.9	A
104	Chongqing	China	41.0	38.5	75.2	71.9	20.3	37.1	A
105	Nizhny Novgorod	Russia	35.7	70.7	62.4	59.2	15.9	15.5	A
106	Suva	Fiji	35.7	45.8	36.1	47.7	30.8	25.1	A
107	Medellin	Colombia	35.0	54.4	28.0	66.3	41.9	19.6	A
108	Cebu City	Philippines	33.2	55.8	38.2	40.4	30.8	17.5	A
109	Kumasi	Ghana	25.1	59.4	35.9	14.9	30.3	17.8	A
110	Kathmandu	Nepal	22.3	42.3	37.6	12.2	21.7	16.6	A
111	Vienna	Austria	58.1	46.8	87.1	74.4	71.1	41.4	A
112	Chengdu	China	40.9	37.8	77.0	71.9	20.5	37.1	A
113	Bogota	Colombia	34.1	48.7	28.0	66.3	41.9	19.6	A
114	Makati	Philippines	33.0	52.1	38.2	40.4	30.8	17.5	A
115	Riga	Latvia	32.4	60.6	-	33.5	76.2	40.3	C

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
116	Aqaba	Jordan	28.9	43.2	66.0	46.5	11.5	17.0	A
117	Merida	Mexico	25.1	53.7	13.3	55.5	40.5	14.1	A
118	Nairobi	Kenya	23.9	48.5	45.1	12.5	28.1	14.5	A
119	Chattogram	Bangladesh	21.7	51.3	48.3	10.8	19.6	15.7	A
120	Venice	Italy	56.8	53.9	84.7	67.8	68.6	34.8	A
121	Guangzhou	China	40.0	38.1	98.6	71.9	19.7	37.1	A
122	Surabaya	Indonesia	32.7	63.7	40.8	35.1	29.9	17.8	A
123	Moscow	Russia	30.5	41.0	62.4	59.2	15.9	15.5	A
124	Amman	Jordan	28.1	40.3	66.0	46.5	11.5	17.0	A
125	Guadalajara	Mexico	24.8	49.9	13.3	55.5	40.5	14.1	A
126	Dhaka	Bangladesh	21.6	49.3	48.3	10.8	19.6	15.7	A
127	Asuncion	Paraguay	17.5	40.4	-	74.7	45.4	14.9	C
128	Dakar	Senegal	11.4	40.5	0.7	10.6	26.3	26.7	A
129	Bologna	Italy	54.0	56.4	88.2	66.9	58.4	29.4	A
130	Hangzhou	China	39.0	33.5	84.0	71.9	20.6	37.1	A
131	Jakarta	Indonesia	32.7	58.6	40.8	35.1	29.9	17.8	A
132	Tbilisi	Georgia	26.7	50.0	-	33.1	40.9	43.1	C
133	Mexico City	Mexico	23.3	39.2	13.3	55.5	40.5	14.1	A
134	Islamabad	Pakistan	10.7	47.5	32.7	8.1	13.0	4.4	A
135	Kampala	Uganda	8.7	31.8	-	68.7	15.1	22.5	C
136	Amsterdam	Netherlands	53.9	37.6	97.6	71.8	65.7	42.8	A
137	Shenzhen	China	38.7	35.3	98.6	71.9	19.7	37.1	A
138	Da Nang	Vietnam	28.2	51.7	-	44.3	28.1	20.8	B
139	Zagreb	Croatia	25.7	61.9	-	17.8	77.4	42.3	C
140	Karachi	Pakistan	10.6	48.0	32.7	8.1	13.0	4.4	A
141	Alexandria	Egypt	2.4	45.0	-	11.3	7.7	30.6	C
142	Paris	France	48.6	27.7	97.9	63.9	69.2	45.1	A
143	Nanjing	China	38.4	52.6	75.3	59.0	7.1	34.5	A
144	Phnom Penh	Cambodia	16.6	43.9	-	18.2	34.5	36.3	C
145	Brno	Czechia	15.5	70.5	-	0.0	84.3	42.4	C
146	Lahore	Pakistan	10.6	48.3	32.7	8.1	13.0	4.4	A
147	Shanghai	China	36.7	31.3	98.6	71.9	19.7	37.1	A
148	Gothenburg	Sweden	31.5	65.3	-	14.5	95.5	63.3	C
149	Ljubljana	Slovenia	10.2	67.0	-	0.0	90.6	33.2	C
150	Valencia	Spain	26.6	59.8	-	31.1	87.1	30.4	C
151	Cork	Ireland	24.5	82.1	-	13.3	91.8	37.4	C
152	Bergen	Norway	21.7	76.4	-	3.2	97.8	48.8	C
153	Antwerp	Belgium	16.9	65.1	-	4.4	87.9	40.4	C
154	Munich	Germany	10.8	18.3	-	27.1	65.0	36.2	C